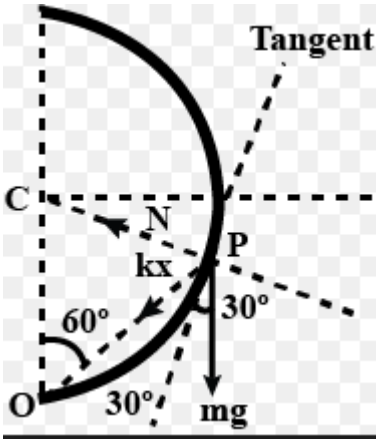
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A smooth semicircular wire track of radius R is fixed in a vertical plane (sec figure). One end of a massless spring of natural length (3/4) R is attached to the lowest point O of the wire track. A small ring of mass m, which can slide on the track, is attached to the other end of the spring. The ring is held stationary of point P such that the spring makes an angle of 60° with the vertical. The spring constant $k = (mg/R)$. Consider the instant when the ring is released. Determine the tangential accelaration of the ring and the normal reaction between ring and track.



(A) $\frac{3}{8}$mg

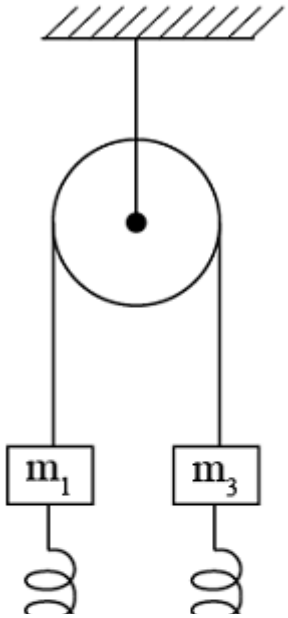
(B) $\frac{6}{5}$mg

(C) $\frac{6}{7}$mg

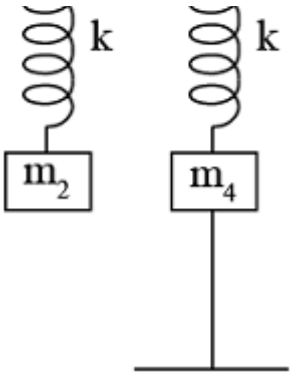
(D) $\frac{8}{3}$mg

Q.2

Four blocks and two springs are arranged as shown in figure, The system at rest, determine the acceleration of all the loads immediately after the lower thread keeping the system in equilibrium has been cut. Assume that the threads are weightless, the mass of the pulley is negligible small, and there is no friction at the point of suspension.



@IITJEE_Advanced



(A) $a_1 = a_2 = a_3 = 0,\ a_4 = \frac{(m_1+m_2-m_3-m_4)g}{m_4}$

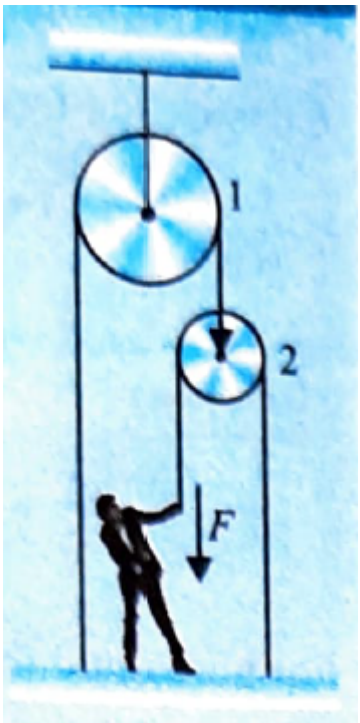
(B) $a_1 = a_2 = a_4 = 0,\ a_3 = \frac{(m_1+m_2-m_3-m_4)g}{m_4}$

(C) $a_1 = a_4 = a_3 = 0,\ a_4 = \frac{(m_1+m_2-m_3-m_4)g}{m_4}$

(D) $a_4 = a_2 = a_3 = 0,\ a_4 = \frac{(m_1+m_2-m_3-m_4)g}{m_4}$

Q.3

A man of mass $60\ \text{kg}$ is standing on a platform of mass $20\ \text{kg}$. With what force F a man pulls on a rope in order to support the platform on which he stands?



(A) 1800 N

(B) 2000 N

(C) 800 N

(D) 10000 N

Q.4

A smooth pulley A of mass M_0 is lying on a frictionless table. A massless rope passes round the pulley and has masses M_1 and M_2 attached to its ends, the two portions of the string being perpendicular to the edge of the table so that the masses hang vertically. Find the acceleration of the pulley.

@IITJEE_Advanced

(A) $\frac{4M_1M_2g}{4M_1M_2+M_0(M_1+M_2)}$

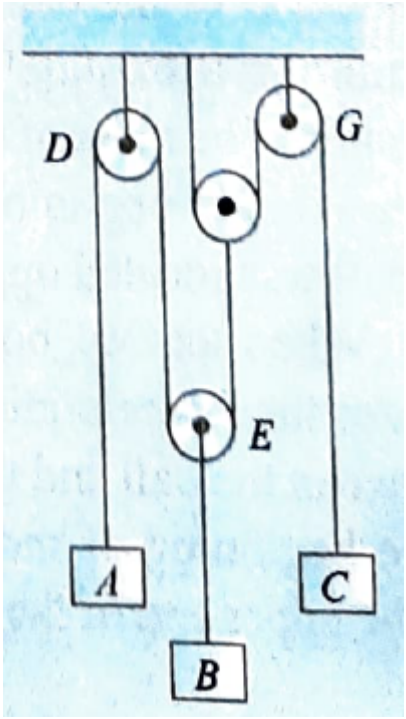
(B) $\frac{5M_1M_2g}{4M_1M_2+M_0(M_1+M_2)}$

(C) $\frac{9M_1M_2g}{4M_1M_2+M_0(M_1+M_2)}$

(D) $\frac{7M_1M_2g}{4M_1M_2+M_0(M_1+M_2)}$

Q.5

Pulleys shown in the system are massless and frictionless. Threads are inextensible. The mass of blocks A, B, and C are $m_1 = 2\ \text{kg}$, $m_2 = 4\ \text{kg}$, and $m_3 = 2.75\ \text{kg}$, respectively. Calculate the acceleration of block c.



(A) $6\ \text{ms}^{-2}$

(B) $8\ \text{ms}^{-2}$

(C) $5\ \text{ms}^{-2}$

(D) $10\ \text{ms}^{-2}$

Q.6

A car is moving in a circular horizontal track of radius $10\ \text{m}$ with a constant speed of 10m/s. A plumb bob is suspended from the roof of the car by a light rigid rod. The angle made by the rod with the track is $(g = 10\ \text{m/s}^2)$

(A) zero

(B) 30°

(C) 45°

(D) 60°

Q.7

@IITJEE_Advanced

One end of string of length l is connected to a particle of mass ' m ' and the other end is connected to a small peg on a smooth horizontal table. If the particle moves in circle with speed 'v' the net force on the particle (directed towards centre) will be
(T represents the tension in the string) :-

(A) $\mathrm{T}+\frac{\mathrm{mv}^2}{l}$

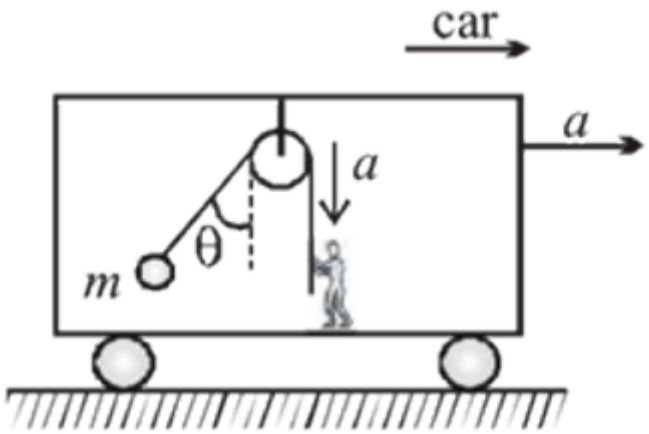
(B) $\mathrm{T}-\frac{\mathrm{mv}^2}{l}$

(C) Zero

(D) T

Q.8

A bob is hanging over a pulley inside a car through a string. The second end of the string
standing in the car. is in the hand of a person
The car is moving with constant acceleration a directed horizontally as shown in figure. Other end of the string is pulled with constant acceleration a vertically. The tension in the string is equal to



(A) $m\sqrt{\mathrm{g}^2+a^2}$

(B) $m\sqrt{\mathrm{g}^2+a^2}-ma$

(C) $m\sqrt{\mathrm{g}^2+a^2}+ma$

(D) $m(\mathrm{g}+a)$

Q.9

Five persons A, B, C, D and E are pulling a cart of mass $100\ \mathrm{kg}$ on a smooth surface and cart is moving with acceleration $3\ \mathrm{m/s^2}$ in east direction. When person A stops pulling, it moves with acceleration $1\ \mathrm{m/s^2}$ in the west direction. When person B stops pulling, it moves with acceleration $24\ \mathrm{m/s^2}$ in the north direction. The magnitude of acceleration of the cart when only A and B pull the cart keeping their directions same as the old directions, is

(A) $26\ \mathrm{m/s^2}$

(B) $3\sqrt{71}\ \mathrm{m/s^2}$

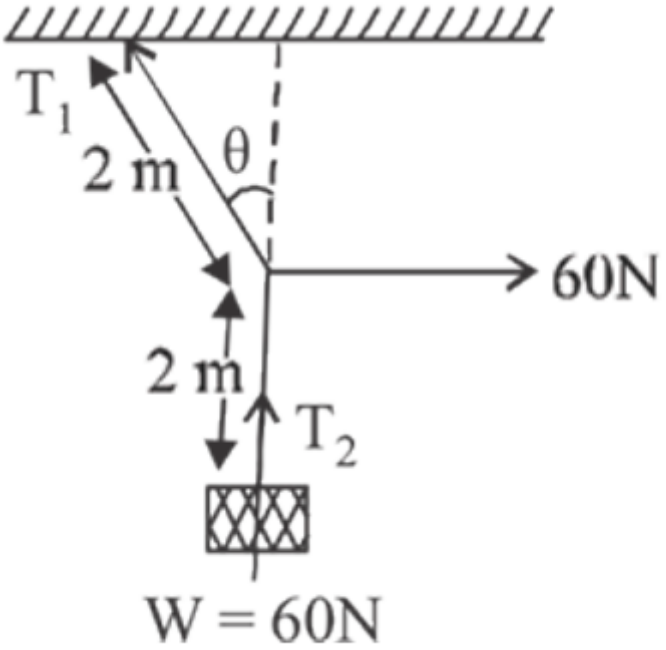
(C) $30\ \mathrm{m/s^2}$

(D) $25\ \mathrm{m/s^2}$

@IITJEE_Advanced

Q.10

For the given situation as shown in the figure, the value of θ to keep the system in equilibrium will be



(A) 30°

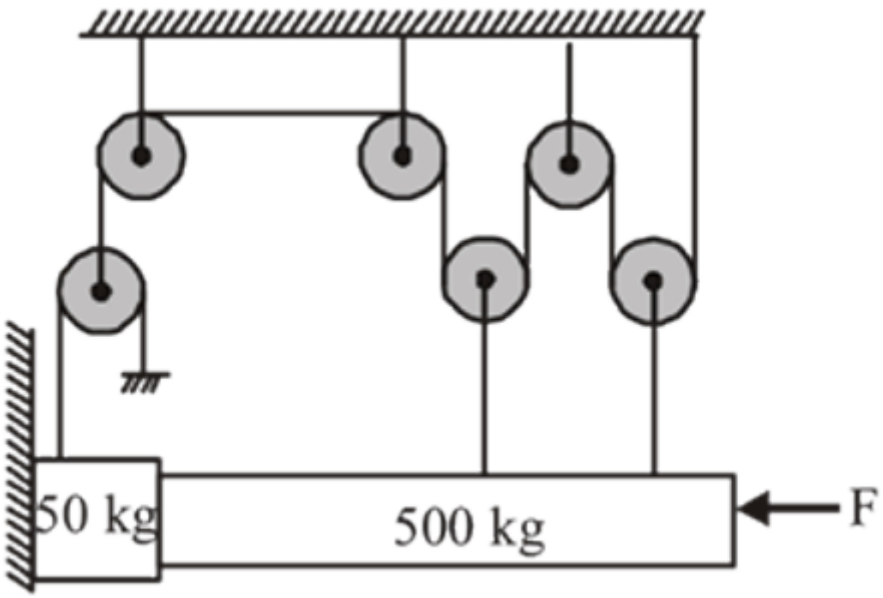
(B) 45°

(C) 0°

(D) 90°

Q.11

If the coefficient of friction between all surfaces is 0.5, then find the minimum force F to have equilibrium of system. (assume strings and pulleys are massless)



(A) $\frac{4000}{17}$ N

(B) $\frac{1000}{17}$ N

@IITJEE_Advanced

(C) $\frac{2000}{17}$ N

(D) $\frac{500}{17}$ N

Q.12

A person of mass $60\ \mathrm{kg}$ is inside a lift of mass 940 kg and presses the button on control panel. The lift starts moving upwards with an acceleration $1.0\ \mathrm{m/s^2}$. If $\mathrm{g} = 10\ \mathrm{ms^{-2}}$, the tension in the supporting cable is

(A) 8600 N

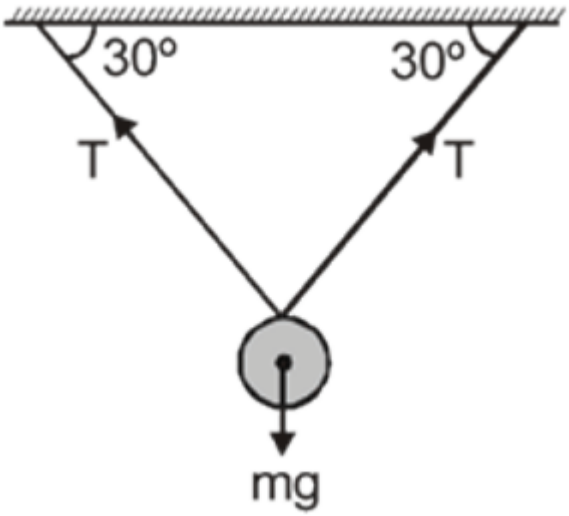
(B) 9680 N

(C) 11000 N

(D) 1200 N

Q.13

A solid sphere of $2\ \mathrm{kg}$ is suspended from a horizontal beam by two supporting wires as shown in fig. Tension in each wire is approximately $\left(\mathrm{g} = 10\ \mathrm{ms^{-2}}\right)$



(A) 30 N

(B) 20 N

(C) 10 N

(D) 5 N

Q.14

A horizontal uniform rope of length L, resting on frictionless horizontal surface, is pulled at one end by force F. What is the tension in the rope at a distance l from the end where the force is applied?

(A) $\mathrm{F}\left(1 - \frac{l}{\mathrm{L}}\right)$

(B) $2\ \mathrm{F}\left(1 - \frac{l}{2\,\mathrm{L}}\right)$

(C) $\frac{\mathrm{F}}{\mathrm{L}}$

@IITJEE_Advanced

(D) $\frac{\mathrm{F}}{l}\left(1-\frac{l}{\mathrm{L}}\right)$

Q.15

A triangular block of mass M with angles 30°, 60°, and 90° rests with its $30^{\circ}-90^{\circ}$ side on a horizontal table. A cubical block of mass m rests on the $60^{\circ}-30^{\circ}$ side. The acceleration which M must have relative to the table to keep m stationary relative to the triangular block assuming frictionless contact is

(A) g

(B) $\frac{\mathrm{g}}{\sqrt{2}}$

(C) $\frac{g}{\sqrt{3}}$

(D) $\frac{\mathrm{g}}{\sqrt{5}}$

Q.16

A monkey of mass $20\ \mathrm{kg}$ is holding a vertical rope. The rope will not break when a mass of $25\ \mathrm{kg}$ is suspended from it but will break if the mass exceeds $25\ \mathrm{kg}$. What is the maximum acceleration with which the monkey can climb up along the rope ? $\left(g=10\ \mathrm{m/s^2}\right)$

(A) $2.5\ \mathrm{m/s}$

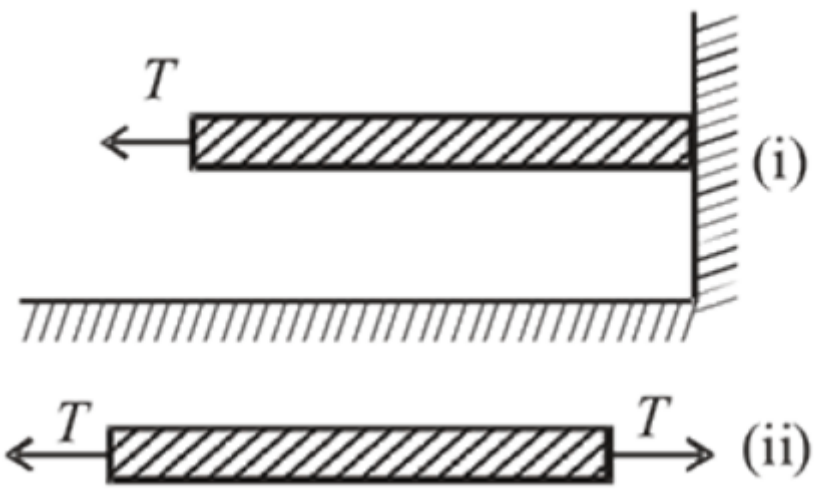
(B) $5\ \mathrm{m/s^2}$

(C) $25\ \mathrm{m/s^2}$

(D) $10\ \mathrm{m/s^2}$

Q.17

In the figure (i) an extensible string is fixed at one end and the other end is pulled by a tension T. In figure (ii) another identical string is pulled by tension T at both the ends. The ratio of elongation in equilibrium of string in (i) to the elongation of string in (ii) is



(A) $1:1$

(B) $1:2$

(C) $2:1$

@IITJEE_Advanced

(D) 0

Q.18

A block of mass m is connected to another block of mass M by a spring (massless) of spring constant k. The block are kept on a smooth horizontal plane. Initially the blocks are at rest and the spring is unstretched. Then a constant force F starts acting on the block of mass M to pull it. Find the force of the block of mass m.

(A) $\frac{MF}{(m+M)}$

(B) $\frac{mF}{M}$

(C) $\frac{(M+m)F}{m}$

(D) $\frac{mF}{(m+M)}$

Q.19

A light spring balance hangs from the hook of the other light spring balance and a block of mass M kg hangs from the former one. Then the true statement about the scale reading is

(A) both the scales read M kg each

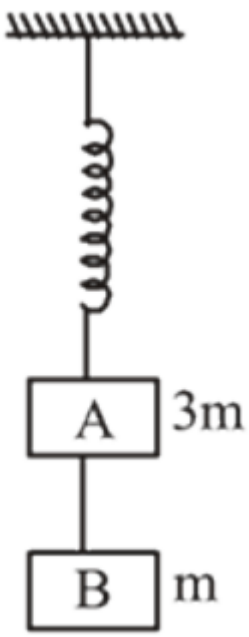
(B) the scale of the lower one reads M kg and of the upper one zero

(C) the reading of the two scales can be anything but the sum of the reading will be M kg

(D) both the scales read $M/2$ kg each

Q.20

Two blocks A and B of masses 3 m and m respectively are connected by a massless and inextensible string. The whole system is suspended by a massless spring as shown in figure. The magnitudes of acceleration of A and B immediately after the string is cut, are respectively:



(A) $\frac{g}{3}$, g

(B) g, g

@IITJEE_Advanced

(C) $\frac{g}{3}, \frac{g}{3}$

(D) $g, \frac{g}{3}$

Answers & Solutions

Q.1 Answer:

$\frac{3}{8}$mg

Solution:

The free-body diagram of the ring is shown in figure. The forces acting on the ring are:

(a) The weight mg acting vertically downwards

(b) Normal force N by the wire track.

Normal force on the ring could be either radially outwards or radially inwards depending on whether the ring presses against the inner surface or outer surface of the track. To ascertain whether normal force is inwards or outwards, assume that, to begin with, it is inwards. Then from $\sum \vec{F} = m\vec{a}$, find the value of normal force. If it is positive. it is inwards and if it is negative, it is outwards.

(c) Force of the spring kx. In the given physical situation, the spring is extended, it will pull the ring. So the spring force kx is along the spring towards O.

Length of the spring in the position shown $= R.(CP = CO = R; \angle COP = \angle OPC = 60^\circ; \angle COP$ **is equilateral** $)$

Change in length of the spring $= R - \left(\frac{3}{4}\right) R = \left(\frac{R}{4}\right)$

$kx = \left(\frac{mg}{R}\right)\left(\frac{R}{4}\right) = \frac{mg}{4}$

Now from $F_t = ma_t$

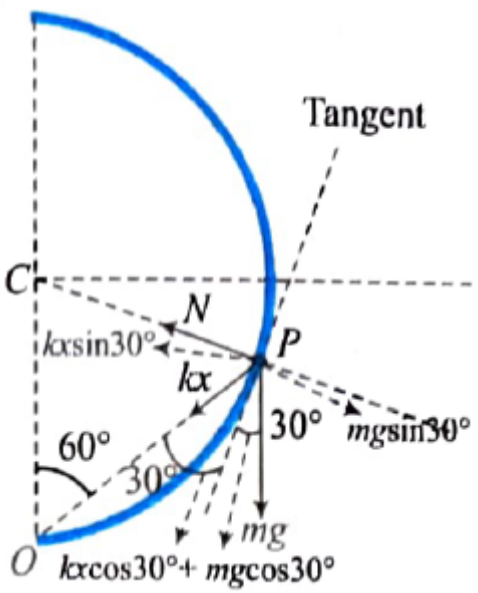
$\left(\frac{mg}{4}\right)\cos 30^\circ + mg\cos 30^\circ = ma_t \Rightarrow a_t = \frac{5\sqrt{3}}{8}$ g

Now consider in radial direction

$N + kx\sin 30^\circ = mg\sin 30^\circ$

$N = mg\sin 30^\circ - kx\sin 30^\circ$

$= mg\left(\frac{1}{2}\right) - \left(\frac{mg}{4}\right)\cdot\left(\frac{1}{2}\right) = \frac{3}{8}mg$



Q.2 Answer:

$a_1 = a_2 = a_3 = 0,\ a_4 = \frac{(m_1+m_2-m_3-m_4)g}{m_4}$

@IITJEE_Advanced

Solution:

For the equilibrium of system of loads,

$(m_1 + m_2) > (m_3 + m_4)$

The force in the left spring, $T_1 = m_2 g$ **Let** T_2 **is the force in the right spring. For the equilibrium of** k m_3**, we have**

$m_3 g + T_2 = T_0 \quad ...(\mathrm{i})$

For $m_1, m_1 g + T_1 = T_0$

As $T_1 = m_2 g$**, therefore**

$m_1 g + m_2 g = T_0$

Substituting this value in (i), we get

$m_3 g + T_2 = (m_1 g + m_2 g)$

or $T_2 = (m_1 + m_2 - m_3)\, g$ **...(ii)**

After cutting, the lower thread, the equations of motion for tre loads are

$m_1 g + T_1 - T_0 = m_1 a_1 \quad ...(\mathrm{iii})$

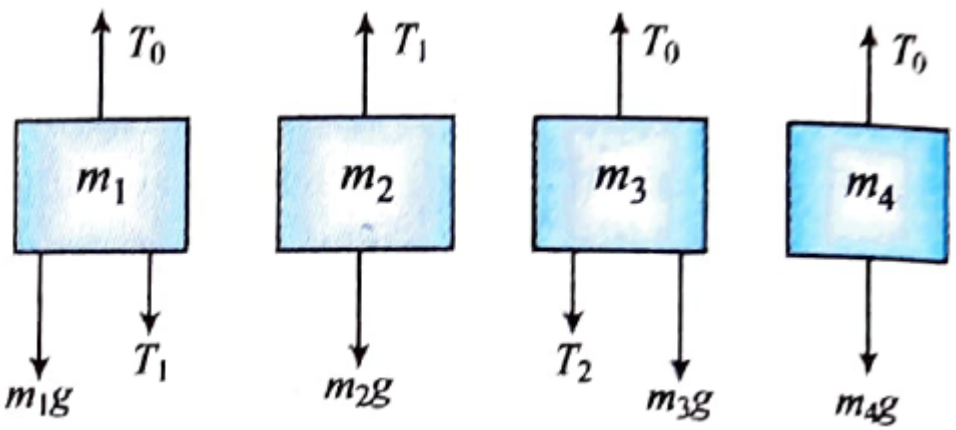
$m_2 g - T_1 = m_2 a_2 \quad ...(\mathrm{iv})$

$T_2 + m_3 g - T_0 = m_3 a_3 \quad ...(\mathrm{v})$

and $T_2 - m_4 g = m_4 a_4 \quad ...(\mathrm{vi})$

Solving above equations, we get

$a_1 = a_2 = a_3 = 0$ and $a_4 = \frac{(m_1 + m_2 - m_3 - m_4) g}{m_4}$



Q.3 Answer:

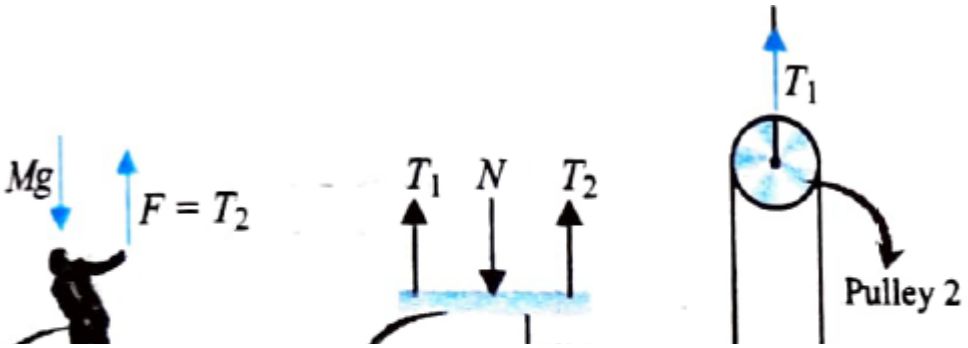
1800 N

Solution:

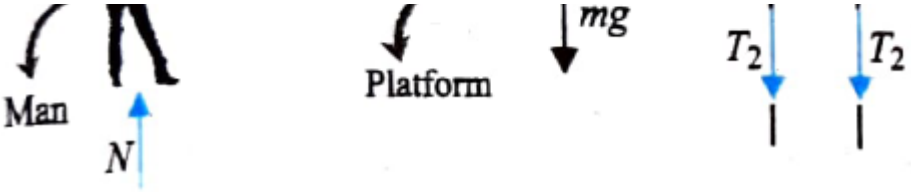
Let N **be the reaction between man and platform.** T_1 **is tension in string passing over pulley 1 and** T_2 **is the tension in sting passing over pulley 2.**

$M =$ **mass of man,** $m =$ **mass of platform**

The free body diagrams of man, platform and pulley 2 are shown in the figure below.



@IITJEE_Advanced



The force exerted by man on platform = Normal reaction exerted by platform on man $= N$

For equilibrium of man: $Mg = T_2 + N$ **...(i)**

For equilibrium of platform: $T_1 + T_2 = N + mg$ **...(ii)**

For equilibrium of pulley $2, T_1 = 2T_2$ **...(iii)**

Adding eqs.(i) and (ii), we get

$$T_1 + T_2 + (T_2 + N) = N + mg + Mg$$
$$T_1 + 2T_2 = (M + m)g$$

Using (iii), we get $2T_2 + 2T_2 = (M + m)g \Rightarrow T_2 = \frac{(M+m)}{4}g$

$$\therefore \quad F = T_2 = \frac{(M+m)}{4}g = \left(\frac{60+20}{4}\right) \times 10 \text{ N} = 200 \text{ N}$$

Now from (i) $N = Mg - T_2 = Mg - F$

$$= Mg - \frac{(M+m)}{4}g = \frac{(3M-m)}{4}g \quad ...(\text{iv})$$
$$\Rightarrow N = \left(\frac{3\times 60-20}{4}\right) \times 10 = 400 \text{ N}$$

From (iv), $mg = 3Mg - 4N$ **Clearly** mg **is maximum when** N **is minimum. Minimum** $N = 0$

So $(mg)_{\max} = 3Mg = 3 \times 60g = 180$ kgwt.

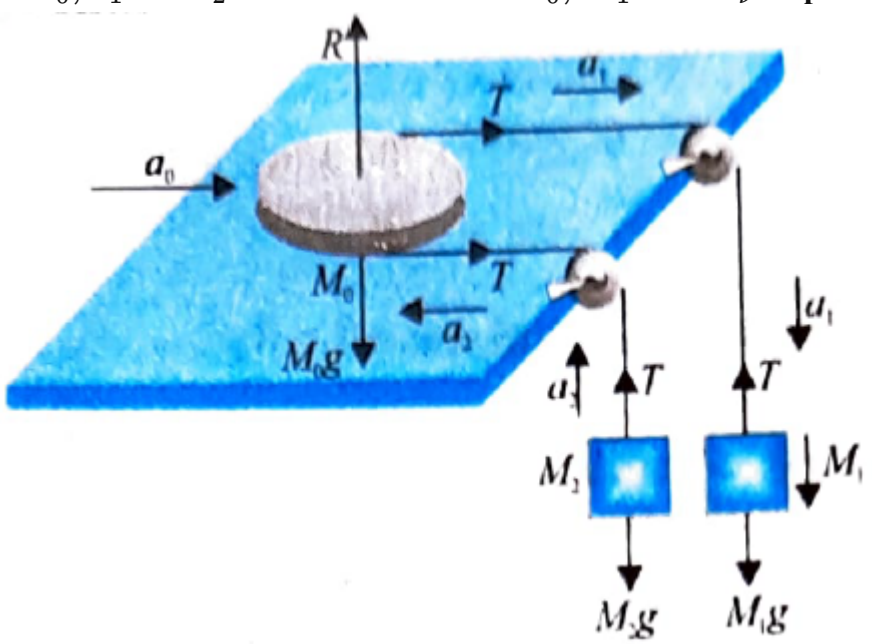
$$(mg)_{\text{mux}} = 3Mg = 3 \times 60 \times 10 = 1800 \text{ N}$$

Q.4 Answer:

$\frac{4M_1M_2g}{4M_1M_2+M_0(M_1+M_2)}$

Solution:

Let a_0, a_1 **and** a_2 **are accelerations of** M_0, M_1 **und** M_ν **respectively. Then**



Constraint relation: $a_0 = \frac{a_1 - a_2}{2} \Rightarrow 2a_0 = a_1 - a.$ **...(i)**

Equation of motion:

@IITJEE_Advanced

For M_0 : $2T = M_0 a_0$ ...(ii)

For M_1 : $M_1 g - T = M_1 a_1$ ...(iii)

For M_2 : $T - M_{-} g = M_2 a_2$ ...(iv)

Substituting values of a_0, a_1 and a_2 from Eqs. (ii), (iii) and (iv) in (i), we get

$$2\left(\frac{2T}{M_0}\right) = \left(g - \frac{T}{M_1}\right) - \left(\frac{T}{M_2} - g\right)$$

$$\frac{4T}{M_0} = 2g - T\left(\frac{1}{M_1} + \frac{1}{M_2}\right)$$

i.e., $\left(\frac{4}{M_0} + \frac{1}{M_1} + \frac{1}{M_2}\right) T = 2g$

This gives $T = \frac{2M_n M_1 M_2 g}{4M_1 M_2 + M_0(M_1 + M_2)}$ ...(v)

Acceleration of pulley of A from eq. (ii),

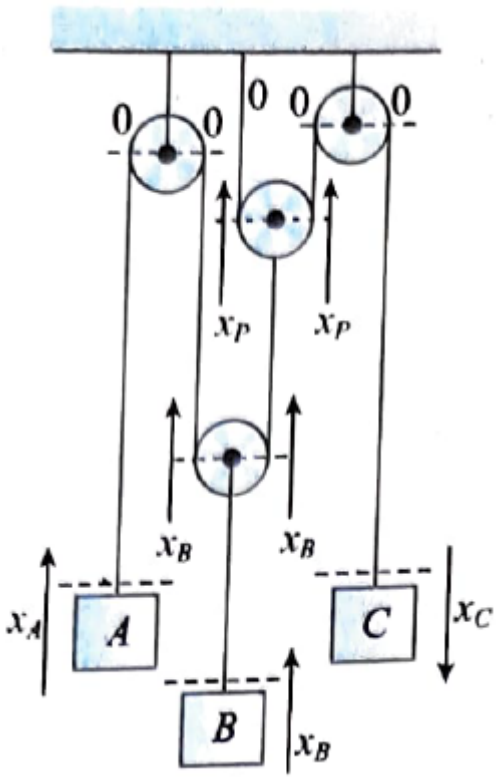
$a_0 = \frac{2T}{M_0} = \frac{4M_1 M_2 g}{4M_1 M_2 + M_0(M_1 + M_2)}$

Q.5 Answer:

$6\ \mathrm{ms}^{-2}$

Solution:

Let the acceleration of blocks A and B be a and b vertically upwards, and acceleration of block C is c downward, respectively. As length of strings are constant, the sum of change in segment lengths should be zero.



For string 1 : [writing from left to right]

$(-x_A) + (-x_B) + (x_P - x_B) = 0$

$x_P = x_A + 2x_B \quad ...(\mathrm{i})$

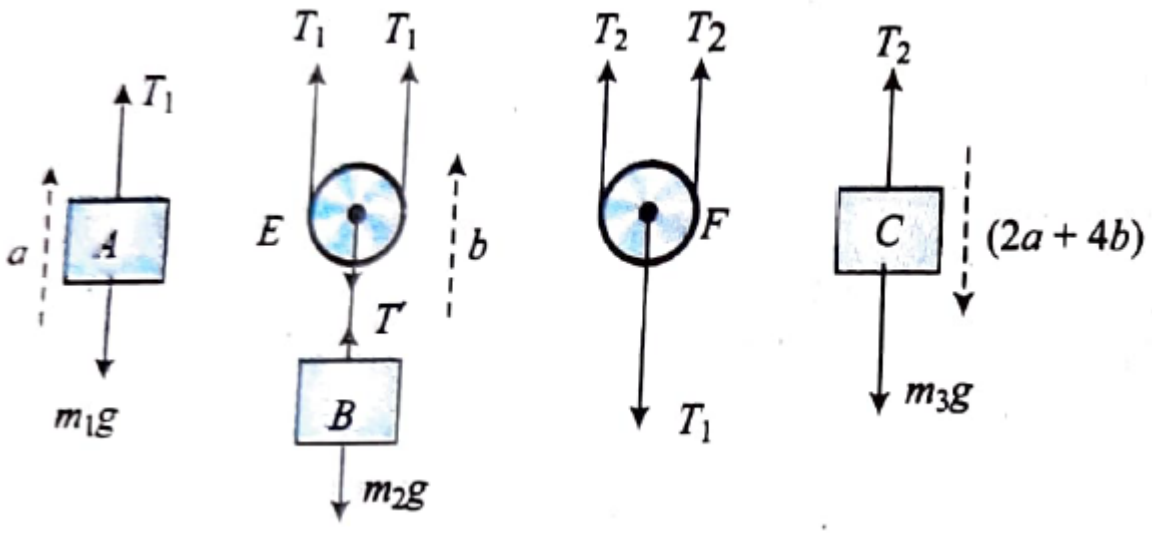
For string 2 : $(-x_p) + (-x_p) + (x_C) = 0$

@IITJEE_Advanced

$x_C = 2x_p$ **...(ii)**

From (i) and (ii): $x_C = 2x_A + 4x_B \Rightarrow \quad c = 2a + 4b$ **...(iii)**

Now considering FBDs, we get the following:



For block A, $\quad T_1 - m_1 g = m_1 a$ **...(i)**

For block B, $\quad 2T_1 - m_2 g = m_2 b$ **...(ii)**

For pulley F, $\quad T_1 = 2T_2$ **...(iii)**

For block C, $\quad m_3 g - T_2 = m_3(2a + 4b)$ **...(iv)**

Solving above equations,

$T_1 = 22\ \text{N}, T_2 = 11\ \text{N}$

$a = 1\ \text{m s}^{-2}, b = 1\ \text{m s}^{-2}$

Hence, acceleration of block A,

$a = 1\ \text{ms}^{-2}(\uparrow)$

Acceleration of block B,

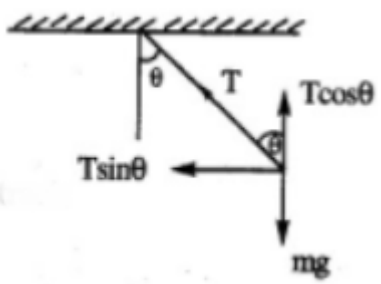
$b = 1\ \text{ms}^{-2}(\uparrow)$

Acceleration of block $C = (2a + 4b) = 6\ \text{m s}^{-2}(\downarrow)$

Q.6 Answer:

45°

Solution:



$T \sin\theta = \frac{mv^2}{R}$

(centripetal force)

$\mathbf{R}$ **= Radius of circular track** $\text{T}\cos\theta = \text{mg}$

@IITJEE_Advanced

$\therefore \tan\theta = \frac{v^2}{Rg}$

or $\tan\theta = \frac{(10)^2}{(10)(10)} = \therefore \theta = 45^\circ$

Q.7 Answer:

T

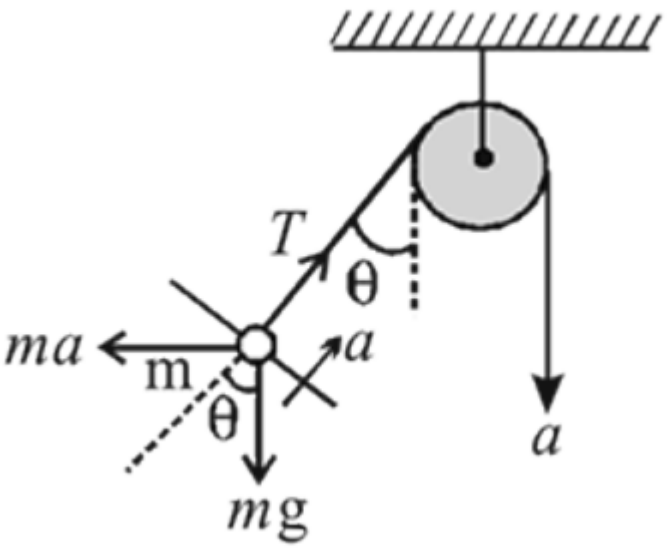
Solution:

Net force on particle in uniform circular motion is centripetal force $\left(\frac{mv^2}{l}\right)$ **which is provided by tension in string so the net force will be equal to tension i.e.,** T.

Q.8 Answer:

$m\sqrt{g^2 + a^2} + ma$

Solution:



(Force diagram in the frame of the car) Applying Newton's law perpendicular to string $mg\sin\theta = ma\cos\theta \quad \Rightarrow \tan\theta = \frac{a}{g}$

Applying Newton's law along string $\Rightarrow T - m\sqrt{g^2 + a^2} = ma$

or $T = m\sqrt{g^2 + a^2} + ma$

Q.9 Answer:

$25\ m/s^2$

Solution:

@IITJEE_Advanced

When all are pulling $\vec{F}_{\text{net}} = 100 \times 3\hat{i}$**...(i)**

When A **stops**

$\vec{F}_{\text{net}} - \vec{F}_A = 100 \times 1(-\hat{i}) \quad \ldots (ii)$

When B **stops**

$\vec{F}_{\text{net}} - \vec{F}_B = 100 \times 24\hat{j}$**...(iii)**

From these three we get $\vec{F}_A + \vec{F}_B = 100(7\hat{i} - 24\hat{j})$

$\therefore$ **Required** $acc^n = \sqrt{7^2 + (-24)^2} = 25 \text{ m/s}^2$

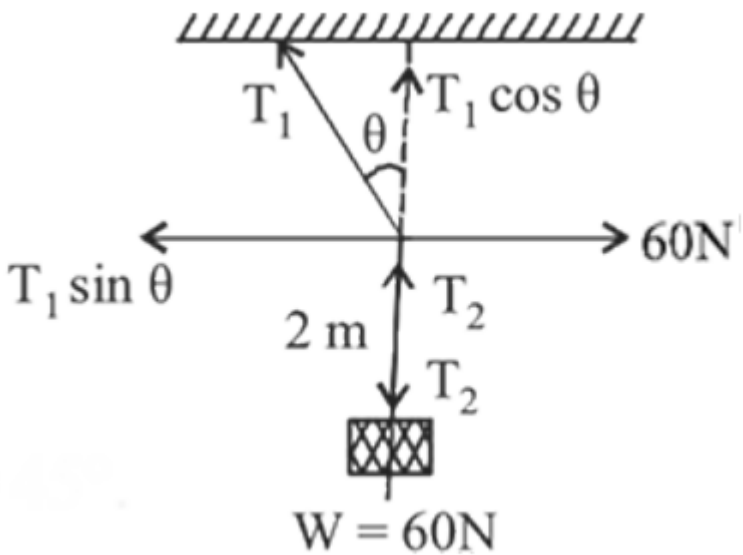
Q.10 Answer:

45°

Solution:

In equilibrium $T_1 \cos\theta = \mathrm{T}_2 = 60 \text{ N}. \quad \ldots (1)$

$\mathrm{T}_1 \sin\theta = 60 \text{ N} \ldots (2)$

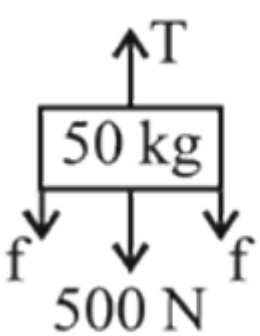


$\therefore \quad \tan\theta = 1 \Rightarrow \quad \theta = 45^\circ.$

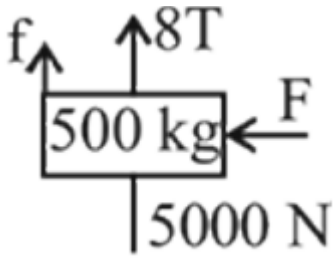
Q.11 Answer:

$\frac{2000}{17}$ N

Solution:



@IITJEE_Advanced



$T = 2f + 500 \quad \ldots$ (i)

$8\,T = 5000 - f \quad \ldots$ (ii)

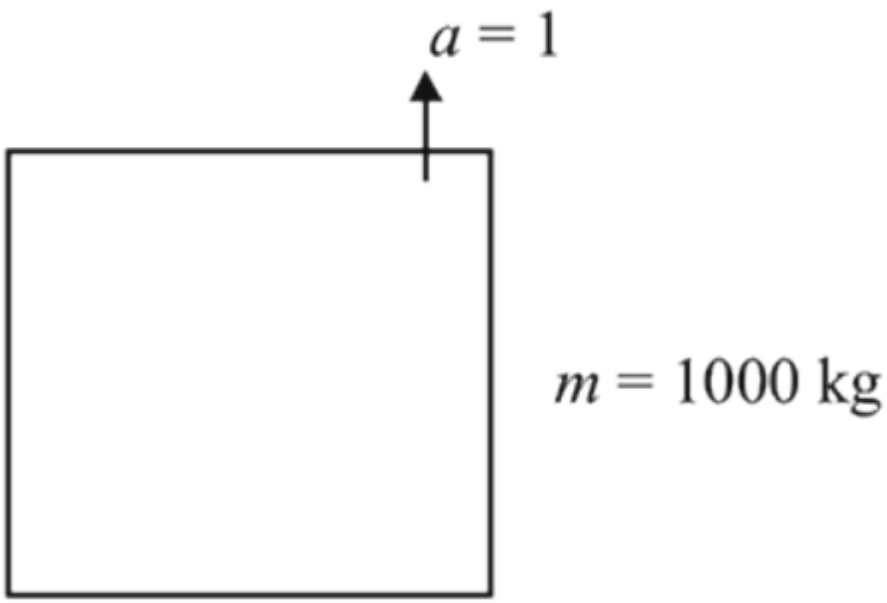
$f = \frac{1}{2}\,F \quad \ldots$**(iii)**

$F = \frac{2000}{17}\ N$

Q.12 Answer:

$11000\ N$

Solution:



Total $\text{mass} = (60 + 940)\text{kg} = 1000\ \text{kg}$

Let T **be the tension in the supporting cable, then** $T - 1000\ g = 1000 \times 1$

$\Rightarrow T = 1000 \times 11 = 11000\ N$

Q.13 Answer:

$20\ N$

Solution:

$2\,T\cos 60^\circ = mg$ **or** $T = mg = 2 \times 10 = 20\ N.$

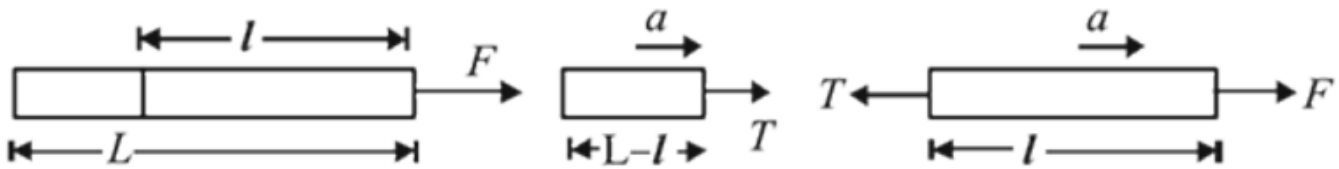
Q.14 Answer:

$F\left(1 - \frac{l}{L}\right)$

Solution:

@IITJEE_Advanced

Let a be the acceleration of the rope and M be its total mass. Then



$T = \frac{M}{L}(L-\ell)a$...(i)

and $F - T = \frac{M}{L} \times \ell a$...(ii)

Dividing (i) and (ii)

$\frac{F-T}{T} = \frac{\ell}{L-\ell} \Rightarrow F(L-\ell) - T(L-\ell) = T\ell$

$\Rightarrow \quad F(L-\ell) = T(L-\ell+\ell) = T \times L$

$\Rightarrow \quad T = F\left(1 - \frac{\ell}{L}\right)$

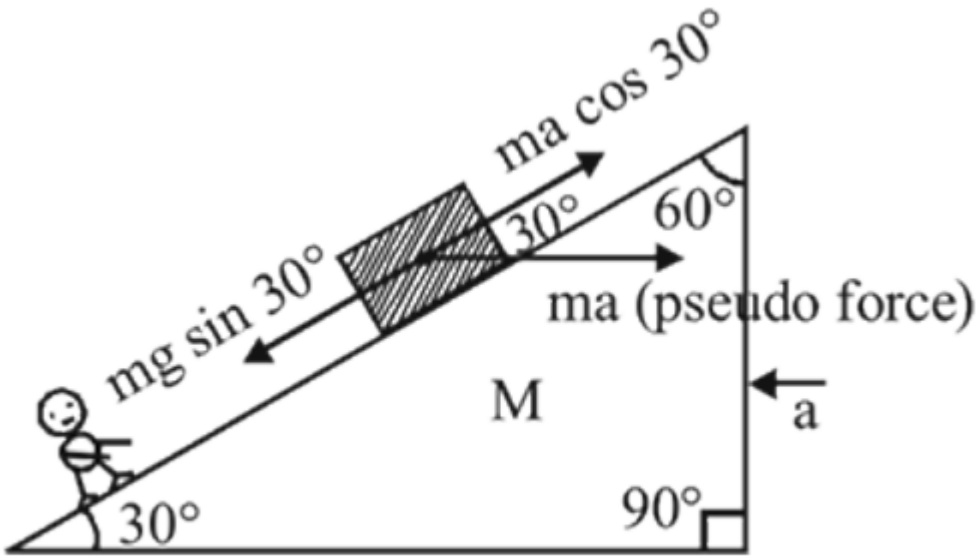
Q.15 Answer:

$\frac{g}{\sqrt{3}}$

Solution:

$\text{macos}\,30^\circ = \text{mg}\sin 30^\circ$

$\therefore \quad \text{a} = \frac{\text{g}}{\sqrt{3}}$



Q.16 Answer:

$2.5\ \text{m/s}$

Solution:

$\text{T} =$ **Tension caused in string by monkey** $= m(g+a)$

$\therefore T \leq 25 \times 10 \Rightarrow 20(10+a) \leq 250$

or, $10 + a12.5 \Rightarrow a \leq 2.5$

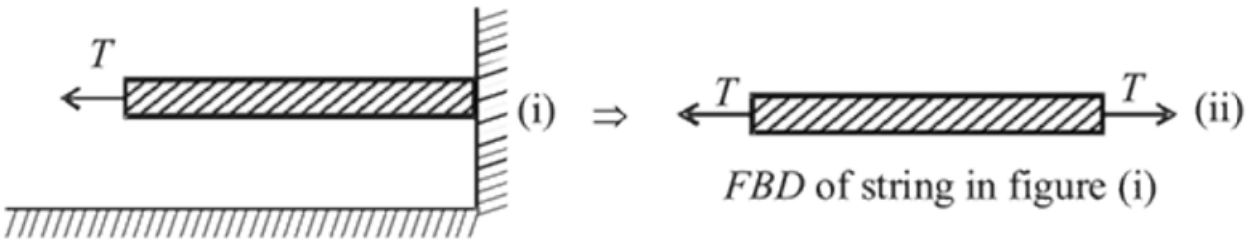
Q.17 Answer:

@IITJEE_Advanced

1 : 1

Solution:

Tension in both string shall be same which can be observed by making FBD of string in figure (i)

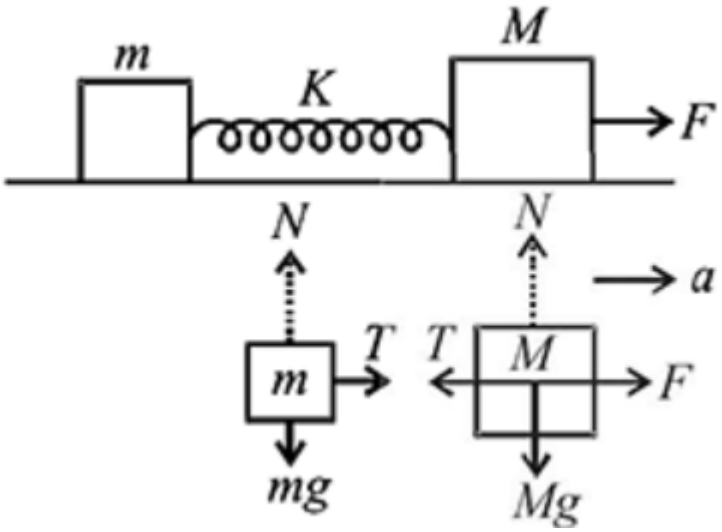


Q.18 Answer:

$\frac{mF}{(m+M)}$

Solution:

Writing free body-diagrams for $m\&M$**,**



we get $T = ma$ **and** $F - T = Ma$ **where** T **is force due to spring** $\Rightarrow F - ma = Ma$ **or,** $F = Ma + ma$

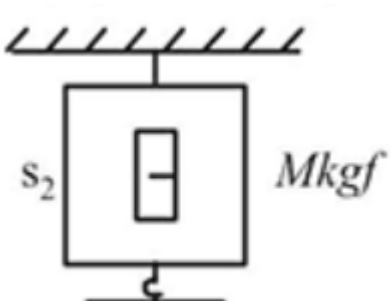
$\therefore \quad a = \frac{F}{M+m}$

Now, force acting on the block of mass m **is** $ma = m\left(\frac{F}{M+m}\right) = \frac{mF}{m+M}$

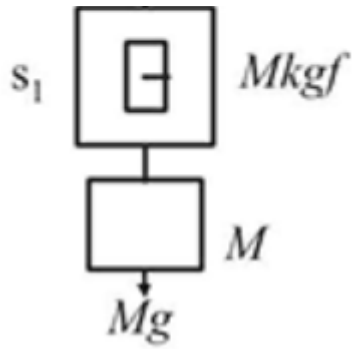
Q.19 Answer:

both the scales read M kg each

Solution:



@IITJEE_Advanced



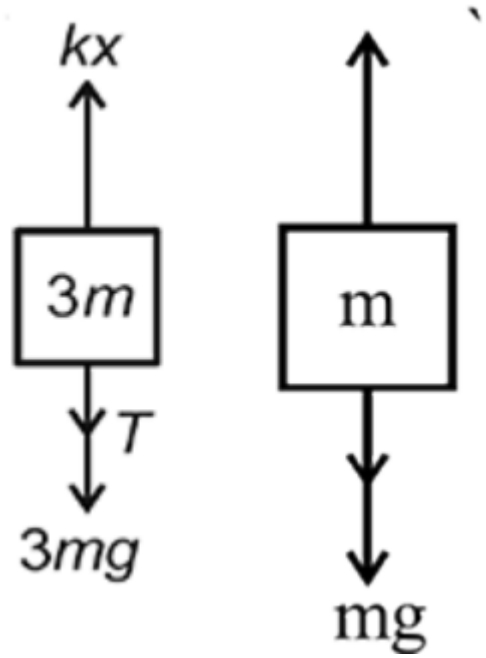
The Earth pulls the block by a force $\angle 1,1/1\ Mg$. The block in turn exerts a force Mg on the spring of spring balance S_1 which therefore shows a reading of M kgf.

The spring S_1 is massless. Therefore it exerts a force of Mg on the spring of spring balance S_2 which shows the reading of Mkgf.

Q.20 Answer:

$\frac{g}{3}$, g

Solution:



Before cutting the string $\mathrm{kx} = \mathrm{T} + 3\mathrm{mg}$

...(i)

$\mathrm{T} = \mathrm{mg}$**...(ii)**

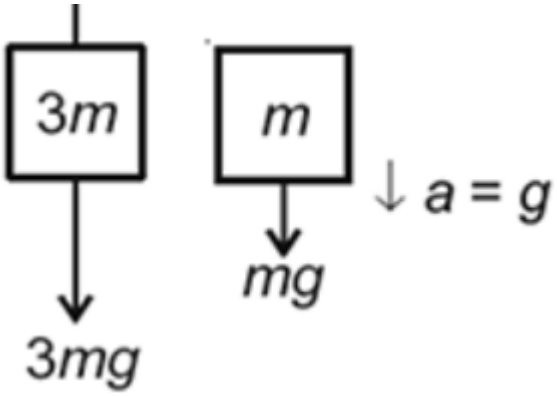
$\Rightarrow \mathrm{kx} = 4\mathrm{mg}$

After cutting the string

$\mathrm{T} = 0$



@IITJEE_Advanced



$a_A = \frac{4mg-3\text{mg}}{3\text{ m}}$

$\mathbf{a_A = \frac{g}{3} \uparrow \text{ and } a_B = \frac{mg}{m} = g \downarrow}$

@IITJEE_Advanced